Linear Programming

1.1 Basic Concepts

Definitions:

Linear Programming (LP, also called Linear optimization) is a method to achieve the best outcome (Max. Profit / Min Cost) in a mathematical model whose requirements are represented by Linear relationship.

Format: Objective + Constraints

Ex.: Maximize: $S_1 \cdot X_1 + S_2 \cdot X_2$ ← Objective

Subject to:

$$\left.\begin{aligned} X_1 + X_2 &\le L \\ F_1 \cdot X_1 + F_2 \cdot X_2 &\le F \\ P_1 \cdot X_1 + P_2 \cdot X_2 &\le P \end{aligned}\right\} \text{Constraints}$$

$$X_1 \geq 0,\ X_2 \geq 0$$

Matrix: maximize $[S_1\ S_2]\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Subject to: $\begin{bmatrix} 1 & 1 \\ F_1 & F_2 \\ P_1 & P_2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \le \begin{bmatrix} L \\ F \\ P \end{bmatrix},\ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Solution: Setting of the Variables

Feasible Solution: Solution that Satisfied all Contraints

Feasible Region: Set of All possible feasible solutions.

Optimal Solution: The Feasible Solution with the largest / Smallest objective Function Value (Maximization / Minimization Problem)

Objective Function Value (OFV): The Value of the objective function correspond to a feasible Solution.

Standard Form

- A linear function to be maximized

  $f(x_1, x_2) = c_1 x_1 + c_2 x_2$

  Max $z = c^T x$

- Problem Constraints of the following form: Subject to:

  $a_{11}x_1 + a_{12}x_2 \leq b_1$

  $a_{21}x_1 + a_{22}x_2 \leq b_2$

  $a_{31}x_1 + a_{32}x_2 \leq b_3$

  $Ax \geq b$

  $x \geq 0$

- Non-negative variables

  $x_1 \geq 0$

  $x_2 \geq 0$

$\max\{c^T x \mid x \in \mathbb{R}^n \wedge Ax \leq b \wedge x \geq 0\}$

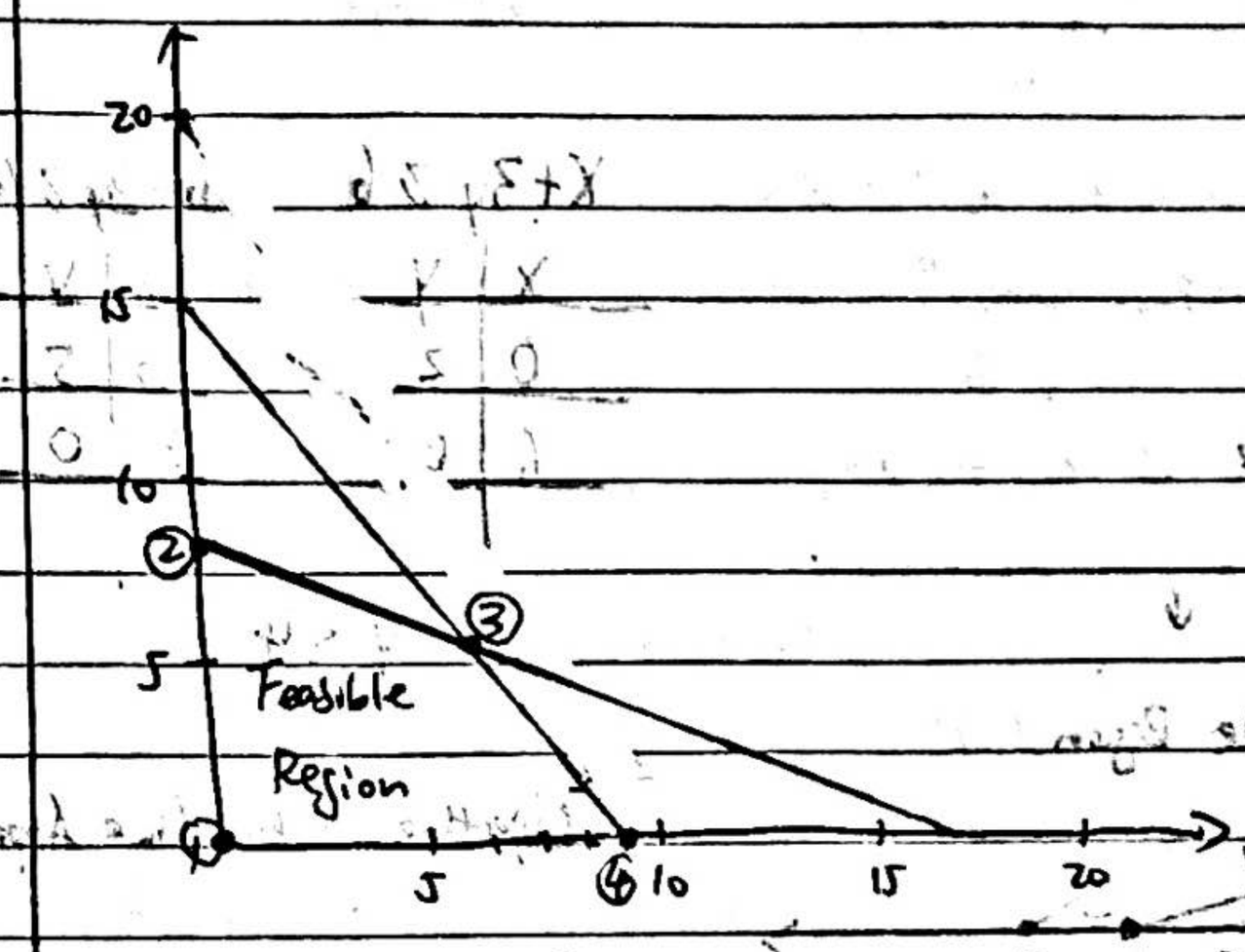
1.2 Maximization — Extreme Corner Points

Max $2X + 5Y$ → Objective Function X, Y → Decision Variables

S.t.

$$\left.\begin{array}{l} X + 2y \leq 16 \;\cdots\; C_1 \\ 5x + 3y \leq 45 \;\cdots\; C_2 \end{array}\right\} \text{Constraints.}$$

$X, Y \geq 0$ → Non-negativity Constraints



$X + 2y \leq 16$

X	Y
0	8
16	0

$5x + 3y \leq 45$

X	Y
0	15
9	0

① $(0,0)$

② $(0,8)$

③ $\longrightarrow \begin{cases} x + 2y = 16 \\ 5x + 3y = 45 \end{cases} \Rightarrow \begin{array}{l} x = 6 \\ y = 5 \end{array}$ $(6,5)$

④ $(9,0)$

Point	$Z = 2x + 5y$
$(0,0)$	0
$(0,8)$	$2(0) + 5(8) = 40$
$(6,5)$	$2(6) + 5(5) = 37$
$(9,0)$	$2(9) + 5(0) = 18$

Optimal Solution
$X = 0,\ Y = 8.$

Maximum OFV (Objective Function Value)
$= 40$

1.3 Graphical Solution – Minimization Problem

Min $Z = 5x + 7y$ — Graphical Method

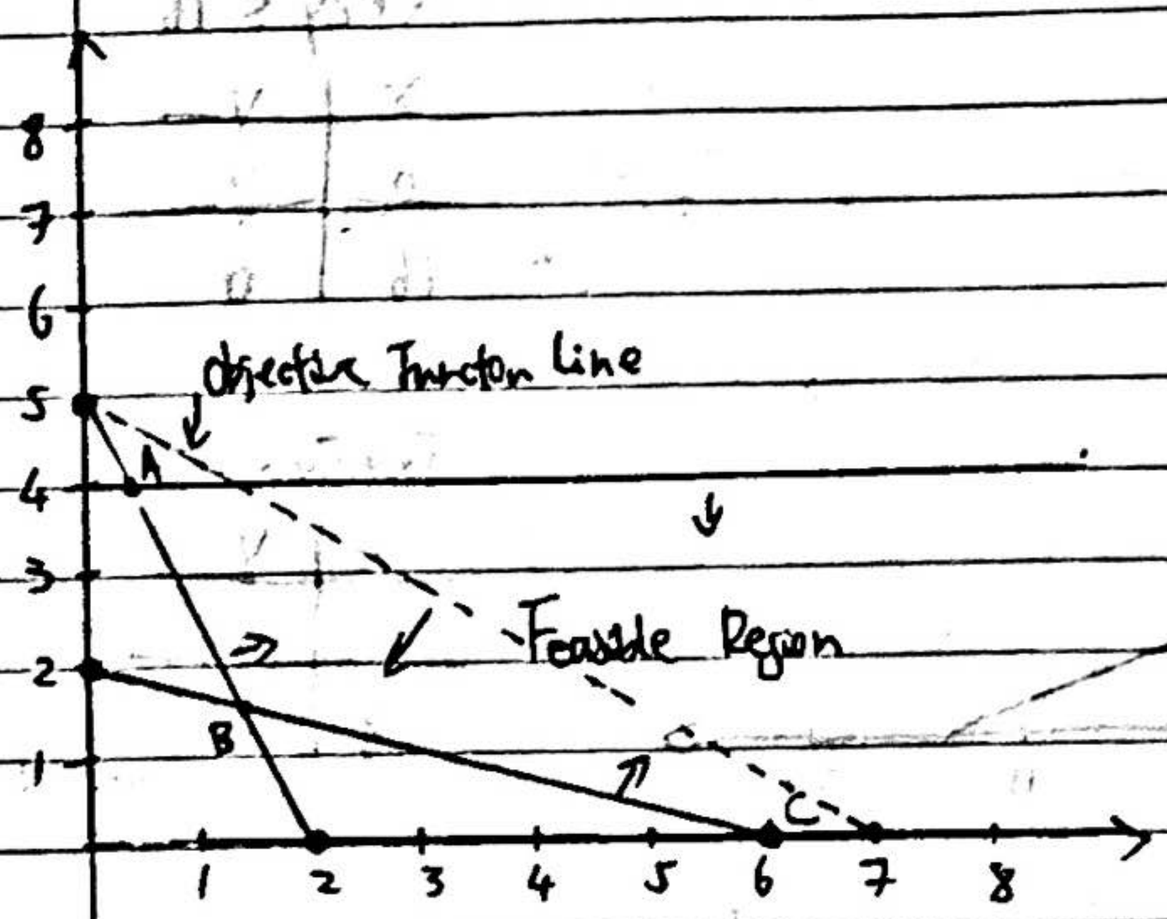
s.t. — 2 Decision Variables Only

$x + 3y \geq 6$ — C_1

$5x + 2y \geq 10$ — C_2

$y \leq 4$ — C_3

$x, y \geq 0$



$x + 3y \geq 6$	
x	y
0	2
6	0

$5x + 2y \geq 10$	
x	y
0	5
2	0

$y \leq 4$

Objective Function Line Approach

Initial OFL: $5x + 7y = 35$

Minimization: OFL Move towards Origin

Maximization: OFL Move away from Origin.

The last corner point touches OFL is the optimal.

Point B: $\begin{cases} x + 3y = 6 \\ 5x + 2y = 10 \end{cases} \Rightarrow (1.38, 1.54)$

Optimal Solution:

$X = 1.38$

$Y = 1.54$

Minimum OFV (Z Value)

$= 17.7$

1.4 Slack/Surplus, Binding Constraints, Standard Form

Max $3x+5y$

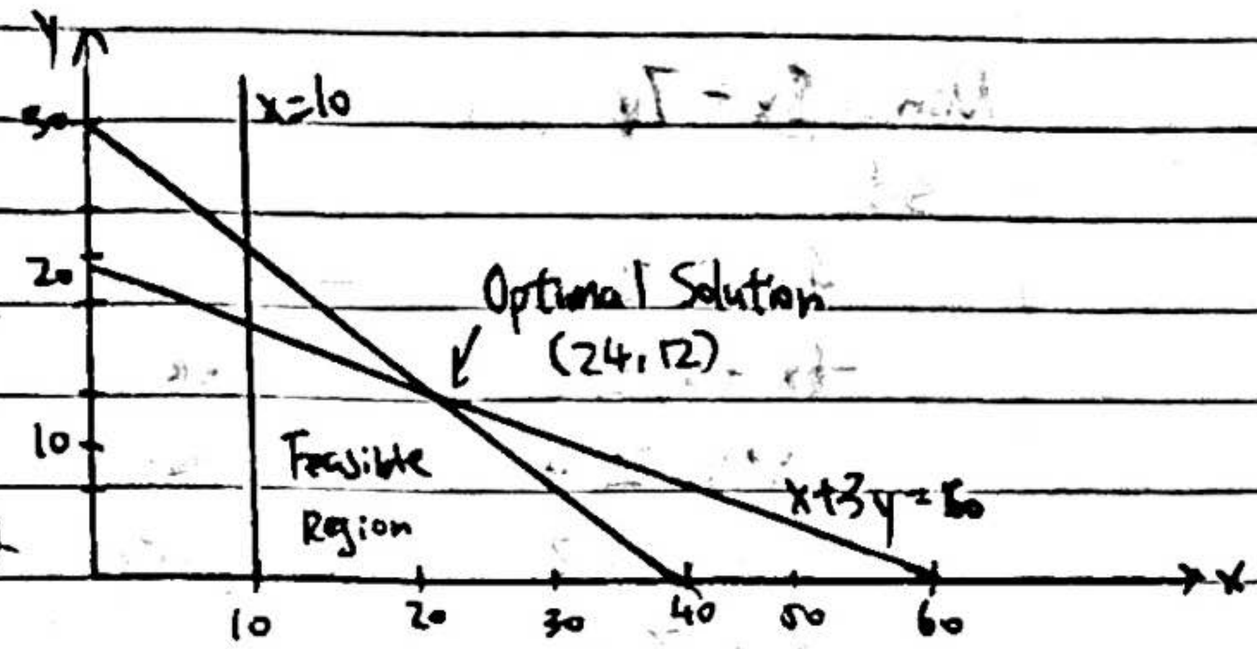
s.t.

$x+3y \le 60$ > Binding

$3x+4y \le 120$

$x \ge 10$ → Non-Binding

$x, y \ge 0$



Optimal Solution $x=24$, $y=12$

Slack = "underutilization" or "left over"

Surplus = "Overutilization" or "exceed over"

Surplus applies to $\ge$ Slack applies to $\le$

$x+3y \le 60$ (Resource 1)

$24+3(12)=60$ No slack Slack $= 60-60=0$

$3x+4y \le 120$ (Resource 2)

$3(24)+4(12)=120$ No slack slack $= 120-120=0$

→ Constraints Binding

Slack $=0$

$x \ge 10$ (At least 10 Units of x)

24 Surplus $= 24-10=14$

Ex. 1 ($\le, \ge \to =$)

Max $3x+5y$

s.t.

| Left side | Operations |

$x+3y \le 60$	→	+ Slack	→	$x+3y+S_1=60$
$3x+4y \le 120$	→	+ Slack	→	$3x+4y+S_2=120$
$x \ge 10$	→	− Surplus	→	$x-S_3=10$
$x, y \ge 0$			→	$x, y, S_1, S_2, S_3 \ge 0$

Standard form

Max $3x+5y+0S_1+0S_2+0S_3$

s.t.

Ex. 2

		Standard Form
Min $8x+7y$		Min $8x+7y+0s_1+0s_2+0s_3$
s.t.		s.t.
$4x+2y \geq 20$	− Surplus	$4x+2y-s_1=20$
$-6x+4y \leq 6$	+ slack	$-6x+4y+s_2=6$
$x+y \geq 4$	− Surplus	$x+y-s_3=4$
$2x-y=2$	0 equality	$2x-y=2$
$x,y \geq 0$		$x,y,s_1,s_2,s_3 \geq 0$

Optimal Solution: $x=3, y=4$

Calculating slack/Surplus

Binding $4x+2y-s_1=20$ $\quad 4(3)+2(4)-s_1=20$

$s_1=0$

Non-Binding $-6x+4y+s_2=6$ $\quad -6(3)+4(4)+s_2=6$

$s_2=8$

Non-Binding $x+y-s_3=4$ $\quad 3+4-s_3=4$

$s_3=3$ (choose s)

Binding $2x-y=2$

1.5 Alternate Solutions, Infeasibility, Unboundedness, Redundancy

Alternate Optimal Solutions

Occur: OFL parrelel to a binding Constraint. (Same slope)

Max $2x + 6y$

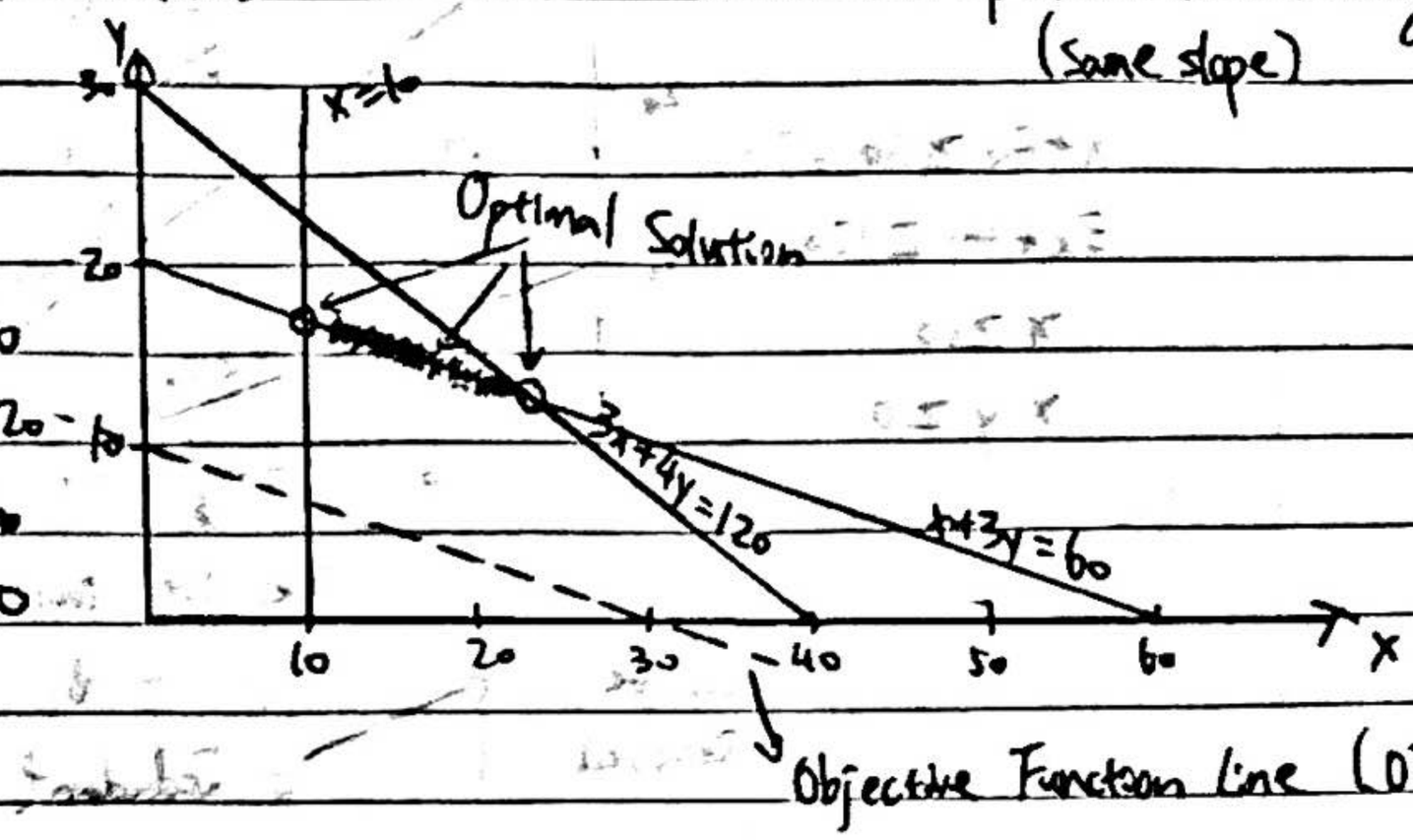
s.t.

$x + 3y \le 60$

$3x + 4y \le 120$

$x \ge 10$

$x, y \ge 0$



Infeasibility

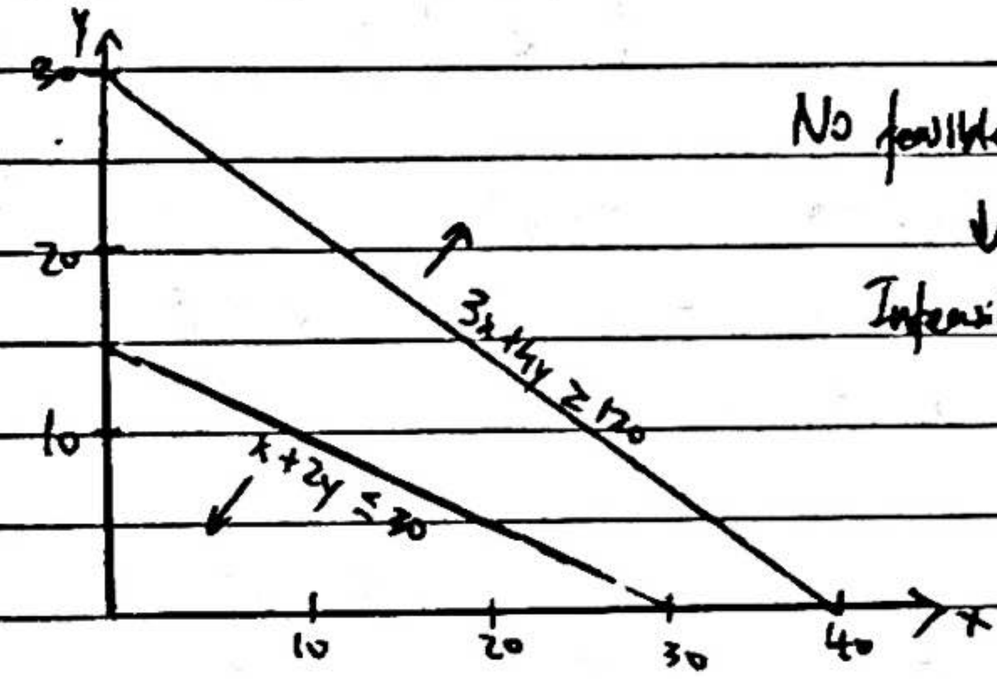
Min $2x + 6y$

s.t.

$x + 2y \le 30$

$3x + 4y \ge 120$

$x, y \ge 0$



No feasible solution

↓

Infeasibility

Unboundedness

Max $2x + 6y$

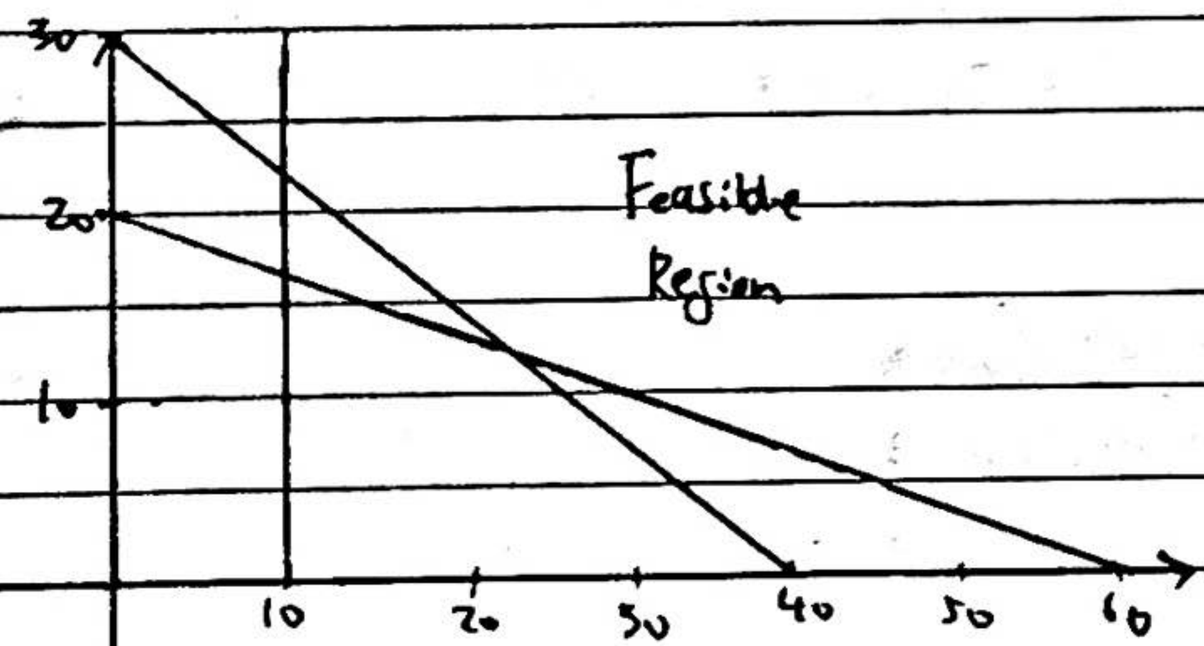
s.t.

$x + 3y \ge 60$

$3x + 4y \ge 120$

$x \ge 10$

$x, y \ge 0$



Maximization problems only

↳ Formulation error

↳ Assuming unlimited supply

Redundancy

Min $4x+5y$

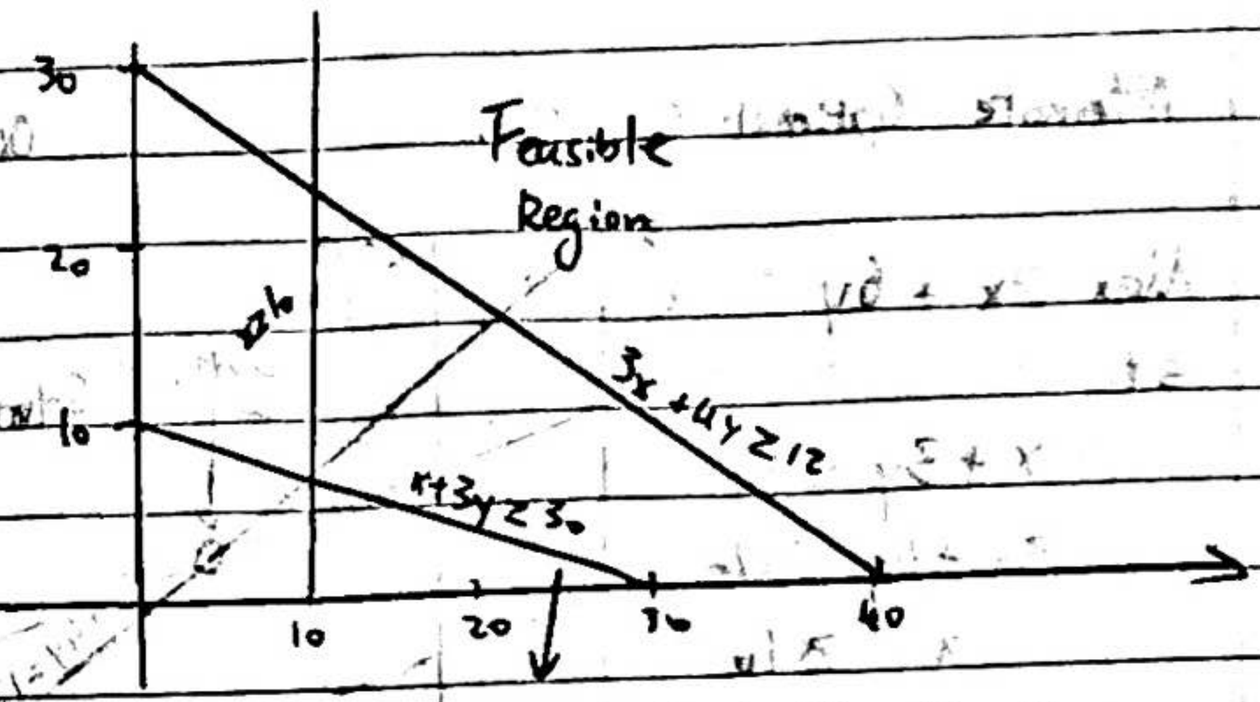
s.t.

$x+3y \geq 30$

$3x+4y \geq 120$

$x \geq 10$

$x, y \geq 0$



Does not touch Feasible Region

Can be Removed

Redudant

1.6 Simplex Method for solving LP

Standard Maximization Problem

Max $z = 8x_1 + 10x_2 + 7x_3 \iff -8x_1 - 10x_2 - 7x_3 + z = 0$

s.t.

$$\left.\begin{aligned} x_1 + 3x_2 + 2x_3 &\le 10 \\ x_1 + 5x_2 + x_3 &\le 8 \end{aligned}\right\} \text{2 slack Variables}$$

$x_1, x_2, x_3 \ge 0$ 3 Variables (non-basic)

x_1	x_2	x_3	s_1	s_2	z		
1	3	2	1	0	0	10	$10/3 = 3\frac{1}{3}$
1	5	1	0	1	0	8	$8/5 = 1(3/5)$
-8	-10	-7	0	0	1	0	

most negative Indicator

$\frac{1}{5}R_2 \to R_2$ *row(1/5, [A], 2) → [A]

$(R_1 = 0)$ $-3R_2 + R_1 \to R_1$ *row+(-3, [A], 2, 1) → [A]

$(R_3 = 0)$ $10R_2 + R_3 \to R_3$ *row+(10, [A], 2, 3) → [A]

x_1	x_2	x_3	s_1	s_2	z		
2/5	0	7/5	1	3/5	0	26/5	$26/5 \div \frac{2}{5} = 13$
1/5	1	1/5	0	1/5	0	8/5	$8/5 \div 1/5 = 8$
-6	0	-5	0	2	1	16	

Most Negative Indicator

$R_2 \to R_2$ *row(5, [A], 2) → [A]

$-2/5R_2 + R_1 \to R_1$ *row+(-2/5, [A], 2, 1) → [A]

$6R_2 + R_3 \to R_3$ *row+(6, [A], 2, 3) → [A]

Non-Basic

x_1	x_2	x_3	s_1	s_2	z	
0	-2	1	1	-1	0	2
1	5	1	0	1	0	8
0	30	1	0	8	1	64

$x_1 = 8, x_2 = 0, x_3 = 0$

$s_1 = 2, s_2 = 0, z = 64$

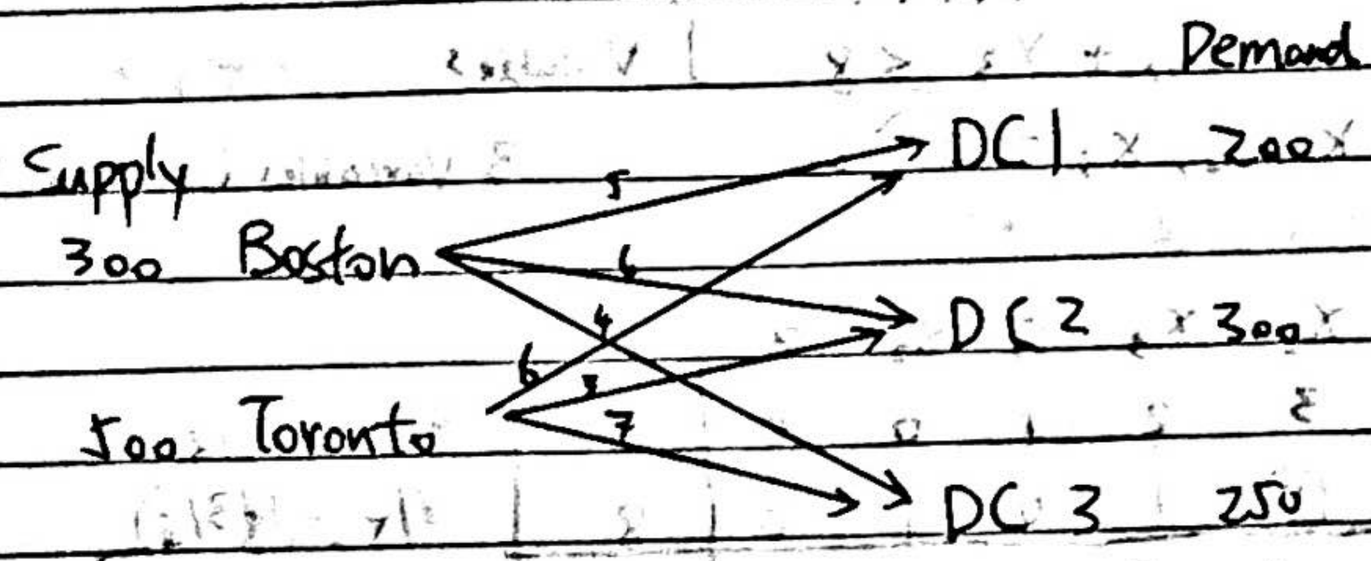
Maximum Value of 64 at (8, 0, 0)

No negative entries. OPTIMAL!

10:

1.7 Transportation Problem

From \ To	DC1	DC2	DC3	Supply
Boston	5	6	4	300
Toronto	6	3	7	500
Demand	200	300	250	



X_{ij} = # units shipped from plant i to DC j

i = B(Boston), T(Toronto) j = 1(DC1), 2(DC2), 3(DC3)

$$\text{Min } 5X_{B_1} + 6X_{B_2} + 4X_{B_3} + 6X_{T_1} + 3X_{T_2} + 7X_{T_3}$$

s.t.

$X_{B_1} + X_{B_2} + X_{B_3} \le 300$ (Boston's Supply)

$X_{T_1} + X_{T_2} + X_{T_3} \le 500$ (Toronto's Supply)

If Tot Supply = Tot Demand

→ Balanced Transportation Problem

$X_{B_1} + X_{T_1} = 200$ (DC1's Demand)

$X_{B_2} + X_{T_2} = 300$ (DC2's Demand)

$X_{B_3} + X_{T_3} = 250$ (DC3's Demand)

$X_{ij} \ge 0$ (Non-Negativity Constraint)